SOLUTION OVERVIEW

CCTV Intelligent Monitoring Platform

AI-powered video surveillance for large-scale, distributed camera deployments

A CCTV monitoring platform that uses AI to watch live camera feeds, detect security and integrity violations, and alert the right people. Every AI detection is checked by a human reviewer before it reaches observers. The platform is proven at national scale, currently monitoring examination centres during major exams, and built to work for any large, distributed camera network.

40,000+
Concurrent camera streams processed in one deployment
PROVEN · NEET 2026

1,00,000
Concurrent camera capacity the platform is built for
ENGINEERED TARGET

16
Detection analytics, each matched to the camera category it applies to
IN PRODUCTION

TECHNOLOGY STACK

Detection Engine	FastAPI (Python) — runs the AI models on every camera feed
Application Backend	Node.js — manages exams, schedules, cameras and alerts
Client Dashboard	React.js — live monitoring for observers and staff
Database	MySQL — stores cameras, detections and reports

AI MODELS IN USE

YOLO26 — Object & Person Detection

Counts people, spots phones and other items, and tracks movement on every camera it processes.

vLLM (Qwen3-VL) — Scene Classification

Classifies what each reachable camera is looking at — Computer Lab, Entry, Server Room — so only the right analytics run on it.

END-TO-END WORKFLOW

From a live camera feed to a verified alert in the observer's hands

1 AUTOMATED

Connect & Check Accessibility

Each camera is connected over its RTSP or HLS link. Offline or unreachable cameras are marked and skipped.

→

2 AI · VLLM

Classify Camera Category

Every accessible camera is classified — Computer Lab, Entry, Server Room, Outside Area and more.

→

3 AI · YOLO26

Scheduled Detection

During each shift, fresh frames are checked by only the analytics that apply to that camera's category.

4 AUTOMATED

Detection Logged

The snapshot is saved with the violation highlighted, plus camera, centre, TC code, shift and time.

→

5 HUMAN VERIFIED

Manual Review

Trained annotators review every detection and filter out false positives before anything goes further.

→

6 DASHBOARD + SMS

Live Alerts to Observers

Verified detections reach the observers' Live Alerts, and an SMS goes to the observers of that centre.

WHAT OBSERVERS CAN DO ON THE DASHBOARD

Live Alerts

Verified detections appear as they are pushed, backed by an SMS to the centre's observers.

Detection Evidence

Every alert opens its snapshot with the violation highlighted, plus camera, centre, TC code and shift.

Search & Filter

Search by centre name or location; filter by exam name, shift timing and analytic type.

Camera Health

Live count of accessible vs. offline cameras per exam and shift, so coverage gaps are visible.

Downloadable Shift Reports

Download a summary report for any shift with every detection, its snapshot and camera details.

Exam & Shift Scheduling

Analytics run only inside each exam's shift windows, with timing and camera categories set per shift.

SUPPORTED CAMERA LINKS

RTSP

Direct Camera & DVR/NVR Streams

Connects straight to IP cameras and DVR/NVR recorders over the standard RTSP streaming protocol.

HLS

Web Streaming Links

Connects to camera feeds published as HLS web streams, such as those served by third-party CCTV portals.

IN PRACTICE — A REAL DETECTION

Phone detected during the RE-NEET examination, 21 June 2026



Red box: the phone on the candidate's desk. **Green box:** the candidate it belongs to. The frame carries the camera's own timestamp as evidence.

HUMAN VERIFIED Reviewed by an annotator and pushed to the observers' Live Alerts.

ANALYTIC
Phone Detection

EXAM
RE-NEET (NTA) · Shift 1

DETECTED
21 June 2026, 02:17 PM

CENTRE
E.M.G Yadava Women's College (Centre A)

LOCATION
Madurai, Tamil Nadu · TC 4106101

CAMERA
Computer Lab · RTSP link

ANALYTICS WE RUN

16 detection analytics in production, each running only on the cameras it applies to

ANALYTIC	WHAT IT FLAGS
EXAM HALL INTEGRITY	
Phone	A mobile phone visible in the exam hall or a secure room.
Close Proximity	Candidates seated or standing too close to each other.
Invigilator	No invigilator in the room, or one standing still in one spot for over 7 minutes.
Crowd < 5	Fewer than 5 candidates present in an exam room.
Zero Person	An exam room found completely empty.
Crowd — Biometric	More than 5 people bunched together in one tight group.
Blind Spot	Part of the camera view blocked, leaving areas unmonitored.
Improper Furniture	Overturned, broken or missing seats in the exam hall.
ENTRY & PERIMETER	
Crowd	More than 10 people gathered at a gate or outside the centre.
Entry Exit	People moving through the entry or exit gate during the exam.
RESTRICTED AREAS	
Intruder	Anyone present in a restricted area during a restricted window, such as before the exam starts.
Server Room Access	People present inside the server room.
Strong Room Movement	Any movement inside the secured strong room.
QP Box	The question-paper box missing from its expected place.
CAMERA INTEGRITY	
Camera Tampering	A camera that has been blocked, covered or blurred.
Camera Inversion	A camera mounted or feeding upside down.